BRIGHTCART FINANCIAL INITIATIVE

E-Commerce Profitability & Marketing Audit

v2.0 Net-Adjusted ELT

Full-Funnel Unit Economics Across 8 Product Categories, 4 Sales Channels & 6 Ad Platforms

BrightCart generated **$277,969.13 in gross revenue** (**$236,318.37 net revenue** after discounts and refunds) across **2,000 orders** over 24 months, yet net profit margins were shrinking. This audit transitions from an initial unadjusted Python pipeline (v1) to a direct MySQL ELT architecture (v2) to isolate true net unit economics and identify a 20% marketing budget reduction.

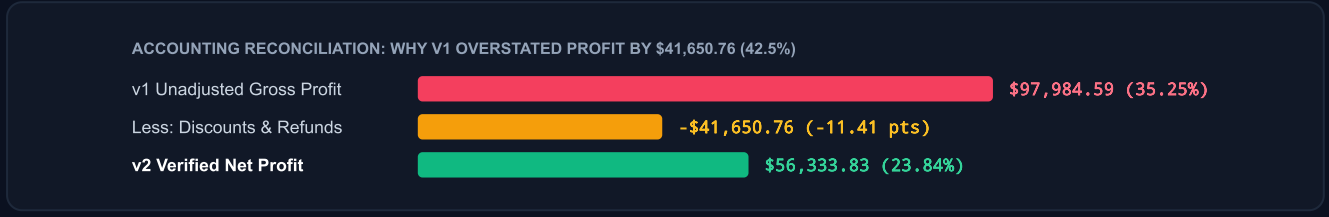
Previous Approach (v1 — Flawed Python Gross Profit Overwrite)

In v1, raw CSVs were pre-processed in Python and `profit` was overwritten as `gross_revenue - total_costs`. That formula stripped out **$19,444.00 in discounts** and **$22,206.76 in returned orders**, artificially inflating total profit to **$97,984.59 (35.25% margin)** and making Mobile App ($54.71) and Website ($54.46) appear tied.

New Approach (v2 — Direct MySQL ELT & Verified Net Accounting)

In v2, raw CSVs were loaded directly into MySQL (`e_commerce_profitability_analysis`) and audited via `00_validation_checks.sql`:

- **Grain & Cost Integrity:** Verified 2,000 unique orders (`COUNT(*) = COUNT(DISTINCT order_id)`) and confirmed `total_costs = product_cost + shipping_cost + transaction_fee + platform_fee` ($179,984.54).
- **Net Revenue & Sunk-Cost Returns:** Confirmed `net_revenue = gross_revenue - (discount_amount + refund_amount)`. For all 144 returned orders, `net_revenue = $0` and `profit = -total_costs`, establishing true net profit at **$56,333.83 (23.84% net margin)**.



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TARGET ENTERPRISE
BrightCart Retail
2,000 Orders | 8 Categories

REVENUE BASELINE
$236,318.37 Net
$277,969.13 Gross Revenue

MARKETING SCOPE
$503,506.14 Spend
144 Platform-Months

QUESTION 01 — CATEGORY PROFITABILITY AUDIT

Category Net Profit Margins & Root-Cause Cost Drivers

SQL View: `vv_category_profitability`

MOST PROFITABLE CATEGORY	LEAST PROFITABLE CATEGORY	BLENDED NET MARGIN
31.13%	**11.94%**	**23.84%**
Electronics ($13,973.46 net profit)	**Books** ($2,249.79 net profit)	Weighted (**22.53%** simple category mean)

CATEGORY	GROSS REV	NET REVENUE	TOTAL COST	NET PROFIT (V2)	NET MARGIN %	V1 GROSS %	SHIP % NET	
1. Electronics	$52,876.49	$44,885.84	$30,912.38	**$13,973.46**	**31.13%**	41.54%	15.91%	4
2. Toys	$38,835.75	$33,604.24	$24,817.99	$8,786.25	**26.15%**	36.52%	21.84%	4
3. Home & Kitchen	$30,683.62	$26,359.75	$19,671.18	$6,688.57	**25.37%**	36.30%	21.88%	4
4. Food & Beverage	$36,085.44	$30,664.28	$23,072.51	$7,591.77	**24.76%**	37.44%	22.97%	4
5. Sports	$37,225.73	$32,324.47	$24,727.84	$7,596.63	**23.50%**	33.96%	21.74%	4
6. Clothing	$37,334.67	$31,375.50	$25,103.02	$6,272.48	**19.99%**	33.07%	25.79%	4
7. Beauty	$22,513.95	$18,256.92	$15,082.04	$3,174.88	**17.39%**	27.75%	31.11%	4
8. Books	$22,413.48	$18,847.37	$16,597.58	**$2,249.79**	**11.94%**	25.95%	**33.23%**	4
Total / Blended	**$277,969.13**	**$236,318.37**	**$179,984.54**	**$56,333.83**	**23.84%**	**35.25%**	**22.84%**	**4**

Root-Cause Driver Attribution: Shipping Explains 90.26% of the 19.19-Point Gap

Using dynamic `DENSE_RANK()` flags (`Highest_profit` vs. `Lowest_profit`), the net margin spread between **Electronics (31.13%)** and **Books (11.94%)** is **19.19 percentage points**:

- **Shipping Cost (+17.32 pts | 90.26% of gap):** Shipping consumes **33.23% of net revenue in Books** vs. **15.91% in Electronics**. High weight-to-price ratios on single books erode unit margins.
- **Product Cost (+2.76 pts):** Books spend **47.52%** of net revenue on product cost vs. **44.76%** in Electronics.
- **Returns (-0.24 pts) & Discounts (-0.89 pts):** Books has slightly lower return rates (**8.37% vs. 8.61%**) and discounts (**6.51% vs. 7.40%**), proving returns and discounts do not drive its low margin.

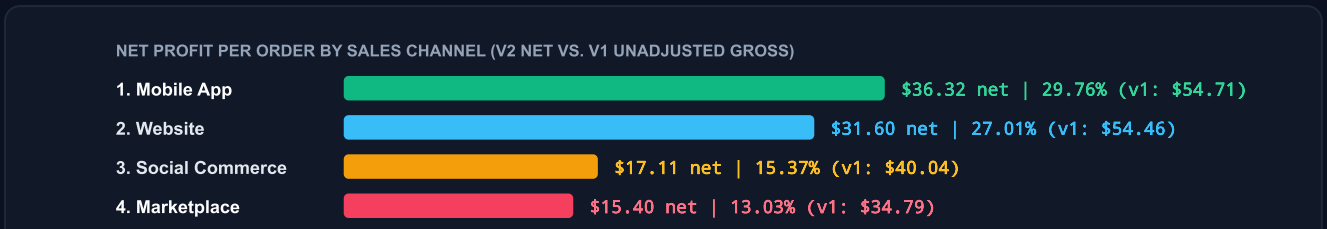
Strategic Recommendation — Category Unit Economics

QUESTION 02 — SALES CHANNEL PROFITABILITY ANALYSIS

Net Profit per Order Across Owned & Third-Party Channels

SQL View: `vv_channel_analysis`

TOP CHANNEL (NET / ORDER)	OWNED DIRECT RUNNER-UP	LOWEST CHANNEL (NET / ORDER)
$36.32	**$31.60**	**$15.40**
Mobile App (29.76% net margin)	**Website** (27.01% net margin)	**Marketplace** (13.03% net margin)



How v2 Net Accounting Changed the Channel Narrative

- **Mobile App Breaks Away from Website (+$4.72/Order):** In v1 unadjusted figures, Mobile App ($54.71, 38.95%) and Website ($54.46, 38.94%) appeared tied—only $0.25 apart. Once discounts and returns are netted out in v2, **Mobile App ($36.32 net profit/order, 29.76% net margin, $122.06 Net AOV, 7.30% return rate)** clearly outperforms **Website ($31.60 net profit/order, 27.01% net margin, $116.97 Net AOV, 7.04% return rate)** by **$4.72 per order** (+2.75 margin points).
- **Marketplace Has the Worst Unit Economics Despite the Lowest Returns: Marketplace** generates just **$15.40 net profit per order (13.03% net margin)** on a strong **$118.15 Net AOV** and the lowest return rate across all channels (**6.44%**). Heavy third-party platform fees cut Marketplace profit per order to less than half of Mobile App (Mobile App is **2.36x more profitable per order**).
- **Social Commerce Suffers a Double Hit: Social Commerce** earns **$17.11 net profit per order (15.37% net margin, $111.32 Net AOV)**, compressed by both platform fees and the highest return rate in the company (**9.14%**).

Strategic Recommendation — Channel Migration

Every repeat buyer migrated from **Marketplace ($15.40)** or **Social Commerce ($17.11)** to the **Mobile App ($36.32)** adds **+$20.92 to +$19.21 in net profit per order** because Owned Channels pay $0 in third-party platform fees. Include QR-code app-exclusive loyalty discounts inside all Marketplace and Social shipments.

QUESTION 03 — RETURN RATE & LOST REVENUE AUDIT

Channel & Category Return Hotspots Across 2,000 Orders

SQL View: `vv_return_rate_by_category_channel`

GROSS REVENUE LOST	TOTAL RETURNED ORDERS	HIGHEST RETURN HOTSPOT
$22,206.76	144 / 2,000	13.33%
7.99% of $277,969.13 gross revenue	**7.20%** overall order return rate	**Social Commerce — Clothing**

RANK (OF 32 CELLS)	SALES CHANNEL	PRODUCT CATEGORY	RETURN RATE %	RELATIVE VISUAL SPREAD	RISK TIER
#1 Highest	Social Commerce	Clothing	13.33%		Critical Hotspot
#2 Highest (Tie)	Social Commerce	Home & Kitchen	13.04%		Critical Hotspot
#2 Highest (Tie)	Social Commerce	Books	13.04%		Critical Hotspot
#4 Highest	Social Commerce	Food & Beverage	12.50%		Critical Hotspot
#5 Highest	Marketplace	Clothing	11.67%		High Sizing Risk
#6 Highest	Website	Electronics	9.90%		Elevated
#7 Highest	Mobile App	Sports	9.47%		Elevated
#8 Highest	Website	Books	9.09%		Elevated
#31 (2nd Lowest)	Marketplace	Sports	1.92%		Low Return
#32 (Lowest)	Marketplace	Food & Beverage	1.61%		Lowest Return

Granular Return Patterns & v2 Sunk-Cost Accounting

- **Social Commerce Concentrates Return Risk:** Social Commerce averages a **9.14% return rate** overall and owns the top four return hotspots out of all 32 channel-category pairs—peaking at **13.33% in Clothing**, **13.04% in Home & Kitchen and Books**, and **12.50% in Food & Beverage** (though it drops to **3.70% in Beauty**).
- **Clothing Sizing Friction Across Channels: Clothing** has the highest category return rate overall (**9.20%**), driven by third-party channels: **13.33% on Social Commerce** and **11.67% on Marketplace** vs. **7.14% on Website** and **7.02% on Mobile App**.
- **v2 Accounting Integration:** Unlike v1 (where the $22,206.76 return figure was reported separately), v2 verifies that all 144 returned orders have `net_revenue = $0` and `profit = -total_costs`, baking their full revenue and fulfillment loss directly into category and channel

QUESTION 04 — MARKETING ROAS & SPEND SCALE AUDIT

Platform Efficiency Ranking & Capital Allocation Mismatch

SQL View: `vv_Marketing_ROI_by_platform`

TOP PLATFORM BY ROAS	LOWEST PLATFORM BY ROAS	24-MONTH BLENDED ROAS
24.02x	**4.81x**	**16.18x**
TikTok Ads ($3.06 CPA | $0.15 CPC)	**Email Marketing** ($17.86 CPA | $0.94 CPC)	$8,146,615.20 rev on $503,506.14 spend

PLATFORM	TOTAL SPEND (V2)	SPEND SHARE	ATTRIBUTED REVENUE (V2)	ROAS	CPA	CPC
1. TikTok Ads	$57,229.22	11.37%	$1,374,627.17	24.02x	$3.06	$0.15
2. Influencer	$97,663.12	19.40%	$2,216,974.26	22.70x	$3.39	$0.17
3. Instagram Ads	$65,154.02	12.94%	$1,024,639.06	15.73x	$4.70	$0.21
4. Google Ads	$152,546.48	30.30%	$2,194,120.83	14.38x	$4.72	$0.25
5. Facebook Ads	$106,451.93	21.14%	$1,218,572.43	11.45x	$6.27	$0.32
6. Email Marketing	$24,461.37	4.86%	$117,681.45	4.81x	$17.86	$0.94
Total / Blended	**$503,506.14**	**100.00%**	**$8,146,615.20**	16.18x	**$4.74**	**$0.23**

Capital Allocation Mismatch Uncovered by Adding Spend Scale in v2

In v1, `vv_Marketing_ROI_by_platform` only output ratios (ROAS, CPA, CPC). By adding `total_spend` and `total_revenue_attributed` in v2, a major budget misallocation is exposed:

- **TikTok Ads Outproduces Facebook on Half the Budget: TikTok Ads (24.02x ROAS, $3.06 CPA)** receives only **11.37% of budget ($57,229.22)** yet generates **$1,374,627.17**—exceeding **Facebook Ads ($1,218,572.43)**, which consumes nearly double the spend (**$106,451.93** at 11.45x ROAS).
- **Legacy Search/Social Dominance: Google Ads ($152.5K)** and **Facebook Ads ($106.5K)** absorb **51.44% of total spend**, while top-performing **TikTok + Influencer** receive just **30.77% of spend** to drive **44.09% of attributed revenue ($3.59M)**.

Are Any Platforms Delivering a Negative Return?

No platform is below **1.0x ROAS** in aggregate; **Email Marketing** is the weakest at **4.81x ROAS** ($17.86 CPA, 5.8x TikTok's CPA). Across all 144 individual platform-months, only **one month loses money below 1.0x**: **Email Marketing in December 2025 (0.67x ROAS, $1,796.66 spend, $163.33 CPA, $3.77 CPC)**.

QUESTION 05 — 20% MARKETING BUDGET OPTIMIZATION STRATEGY

SQL: WHERE cumulative_pct_of_budget < 21

Cumulative Window-Function Cut of 34 Lowest-ROAS Months

RECOMMENDED SPEND CUT
$104,532.02
20.76% of $503,506.14 total budget

CAMPAIGN MONTHS CUT
34 / 144
ROAS range: **0.67x to 6.60x**

ATTRIBUTED REVENUE IMPACT
-5.85% Only
Retained budget ROAS jumps **16.18x → 19.22x**

PLATFORM	MONTHS CUT	SPEND CUT ($)	SHARE OF CUT	% OF PLATFORM CUT	ROAS RANGE IN CUT	AVG CPA / CPC
Facebook Ads	7	**$36,589.02**	**35.00%**	34.37%	2.51x – 6.60x	$12.23 / $0.44
Google Ads	4	**$20,919.54**	**20.01%**	13.71%	3.32x – 4.27x	$13.70 / $0.35
Influencer	4	$16,985.14	16.25%	17.39%	3.78x – 5.83x	$10.53 / $0.25
Email Marketing	**15**	**$16,927.46**	16.19%	**69.20%**	0.67x – 6.57x	$35.07 / $1.34
Instagram Ads	2	$8,696.44	8.32%	13.35%	5.19x – 5.76x	$10.12 / $0.26
TikTok Ads	2	$4,414.42	4.22%	7.71%	6.01x – 6.28x	$8.55 / $0.17
Total 34-Month Cut	**34**	**$104,532.02**	**100.00%**	**20.76% of Budget**	**0.67x – 6.60x**	**$22.17 / $0.78**

How the SQL Window Function Isolates the Optimal 20% Cut

- **Precision Window Framing:** Using `SUM(spend) OVER (ORDER BY roas ASC ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)` inside the `ranked_spend` CTE prevents tied-ROAS merging (at 3.89x and 4.97x) and stops cleanly at 34 rows (**$104,532.02 / 20.76%**). Adding the 35th row (TikTok Jul 2025, 6.63x ROAS, $3,326.90) would push the cut to 21.42%.
- **Frequency vs. Dollar Volume: Email Marketing** accounts for the most months cut (**15 of 24 months, or 69.20% of Email's budget**), but because Email is only 4.86% of total spend, **Facebook Ads ($36,589.02)** and **Google Ads ($20,919.54)** provide **55.01% of total dollar savings**.

Bottom 5 Individual Platform-Months in the Cut Set (Verified CPA & CPC)

1. **Email Marketing (2025-12):** 0.67x ROAS | $1,796.66 spend | CPA **$163.33** | CPC **$3.77** (0.36% cum. budget)
2. **Email Marketing (2025-10):** 1.21x ROAS | $1,398.97 spend | CPA **$58.29** | CPC **$1.69**

EXECUTIVE SUMMARY & ACTION PLAN

Top 3 Recommendations to Restore BrightCart Net Margins

CEO Decision Summary

Recommendation 01 — Cut 34 Low-ROAS Ad Months & Reallocate to TikTok / Influencer

$104,532.02 Saved (20.76%)

Suspend or cap monthly spend across the **34 lowest-ROAS platform-months (≤6.60x ROAS)** to trim **$104,532.02 (20.76% of budget)** while sacrificing only **5.85% of attributed revenue ($476.2K)**—lifting blended ROAS on the remaining $399.0K budget from **16.18x to 19.22x**:

- **Target High-Dollar Troughs:** Enforce monthly ROAS circuit-breakers on **Facebook Ads** (7 months cut, saving **$36,589.02**) and **Google Ads** (4 months cut, saving **$20,919.54**), which together supply **55.01%** of the budget cut.
- **Restructure Email Marketing:** Cut **69.20% of Email spend ($16,927.46 across 15 months)**, eliminating the money-losing December 2025 campaign (0.67x ROAS, $163.33 CPA), and redeploy growth capital into **TikTok Ads (24.02x ROAS, $3.06 CPA)** and **Influencer Campaigns (22.70x ROAS, $3.39 CPA)**.

Recommendation 02 — Enforce Free-Shipping Thresholds on Books & Beauty

+17.32 pt Shipping Gap

Shipping costs consume **33.23% of net revenue in Books (11.94% net margin)** and **31.11% in Beauty (17.39% net margin)**, accounting for **90.26% of the 19.19-point margin gap** against **Electronics (31.13% net margin, 15.91% shipping)**:

- **Minimum Order Value (MOV) Gate:** Set a **$35+ minimum order threshold for free shipping** on Books and Beauty orders, passing a flat $4.99 shipping fee onto single-item low-basket purchases.
- **Multi-Unit Product Bundling:** Introduce multi-book sets and skincare regimens to raise category AOV above $35 and dilute per-order fulfillment cost from 33.2% toward the 22.8% company average.

Recommendation 03 — Migrate Repeat Buyers to Mobile App & Curb Clothing Returns

+$20.92 Net Profit / Order

Mobile App ($36.32 net profit/order, 29.76% net margin) generates **2.36x more net profit per order** than **Marketplace ($15.40, 13.03% margin)** and **2.12x more** than **Social Commerce ($17.11, 15.37% margin)**:

- **Zero-Fee Channel Migration:** Use package inserts with QR-code loyalty rewards to convert first-time Marketplace and Social buyers into repeat Mobile App customers, capturing **+$19.21 to +$20.92 in incremental net profit per order**.
- **Fix Third-Party Apparel Sizing:** Deploy mandatory fit predictors and sizing charts on **Social Commerce Clothing (13.33% return rate)** and **Marketplace Clothing (11.67%)** to directly reduce the **$22,206.76 (7.99% of gross revenue)** lost to returns.